

PROSIT: AI-POWERED PERSONAL RESOURCE OPTIMIZATION & SUBSCRIPTION INTELLIGENCE SYSTEM

Sprint 1 Evaluation – Software Engineering (22AIE311)

Group Number: 7

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INTRODUCTION & PROJECT OVERVIEW

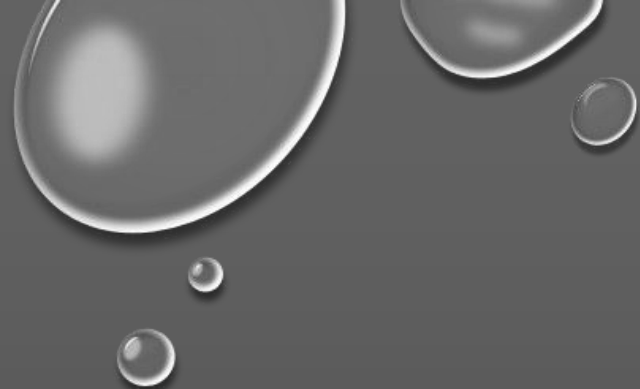
- **CONCEPT:** A CENTRALIZED, PRIVACY-FOCUSED WEB PLATFORM FOR MANAGING DIGITAL SUBSCRIPTIONS AND RECURRING BILLS.
 - **CORE OBJECTIVE:** TO HELP USERS RECLAIM FINANCIAL CONTROL AND OPTIMIZE DIGITAL RESOURCE USAGE.
 - **UNIQUE SELLING POINT:** UNLIKE TRADITIONAL APPS, PROSIT TRACKS **ACTUAL USAGE TIME** (VIA DEVICE AGENTS) TO CALCULATE THE REAL RETURN ON INVESTMENT (ROI) FOR EACH SUBSCRIPTION.
 - **PRIVACY:** A "MANUAL-FIRST" APPROACH THAT AVOIDS RISKY DIRECT BANK INTEGRATIONS.
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DRAWBACKS OF EXISTING SYSTEMS

- **FRAGMENTED MONITORING:** SUBSCRIPTIONS AND BILLS ARE SPREAD ACROSS MULTIPLE PLATFORMS, MAKING IT HARD TO TRACK TOTAL SPEND.
- **PRIVACY & SECURITY:** DEPENDENCY ON BANK API INTEGRATIONS INTRODUCES DATA VULNERABILITY AND REGULATORY RISKS.
- **LACK OF INSIGHT:** MOST APPS SHOW WHAT YOU PAY, BUT NOT HOW MUCH YOU USE THE SERVICE.
- **SUBSCRIPTION FATIGUE:** NO AUTOMATED WAY TO IDENTIFY "GHOST" SUBSCRIPTIONS THAT ARE RARELY USED BUT STILL BILLED.

- **FRAMEWORK:** AGILE DEVELOPMENT PROCESS (CURRENTLY IN ELABORATION PHASE).
- **FRONTEND:** REACT.JS / NEXT.JS WITH TAILWIND CSS FOR A RESPONSIVE, INTERACTIVE DASHBOARD.
- **BACKEND:** NODE.JS (EXPRESS.JS) FOR HIGH-PERFORMANCE API HANDLING AND BACKGROUND SCHEDULING.
- **DATA COLLECTION:** CUSTOM WINDOWS AND ANDROID AGENTS FOR NON-INTRUSIVE USAGE TRACKING.
- **DATABASE:** POSTGRESQL/MONGODB FOR SECURE, SCALABLE DATA STORAGE.

METHODOLOGY & TECH STACK

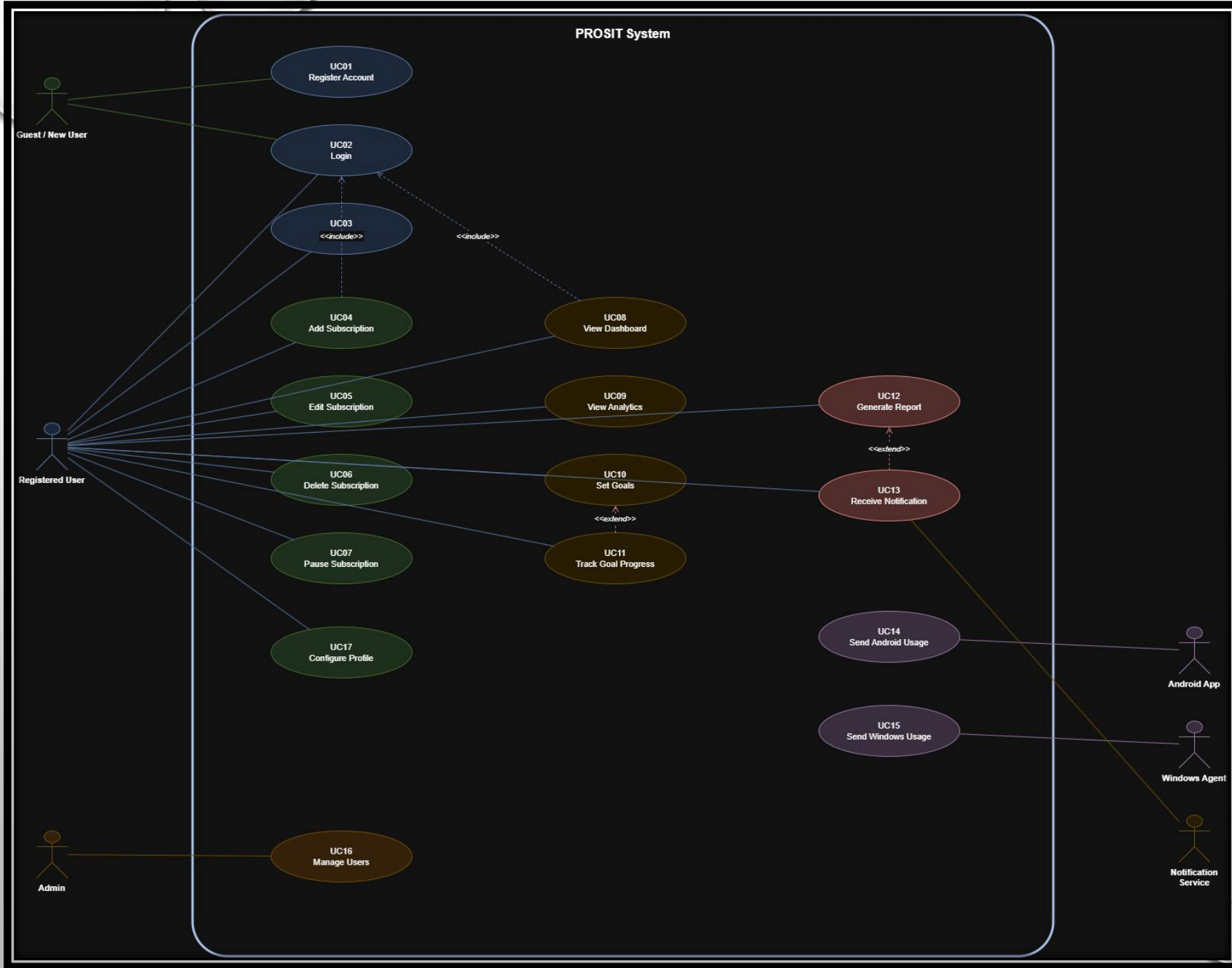


USE CASE MODELING (UML)

- **PRIMARY ACTOR:** REGISTERED USER (MANAGES SUBSCRIPTIONS, VIEWS ANALYTICS, SETS GOALS).
- **SYSTEM ACTORS:** ANDROID/WINDOWS AGENTS (AUTOMATIC USAGE DATA PUSHES).
- **EXTERNAL SERVICE:** NOTIFICATION SERVICE (FCM/SENDGRID FOR ALERTS).

CRITICAL USE CASES:

- MANAGE SUBSCRIPTIONS (CRUD)
- MONITOR RESOURCE USAGE
- GENERATE AI-DRIVEN ROI ANALYTICS
- CONFIGURE PAYMENT REMINDERS

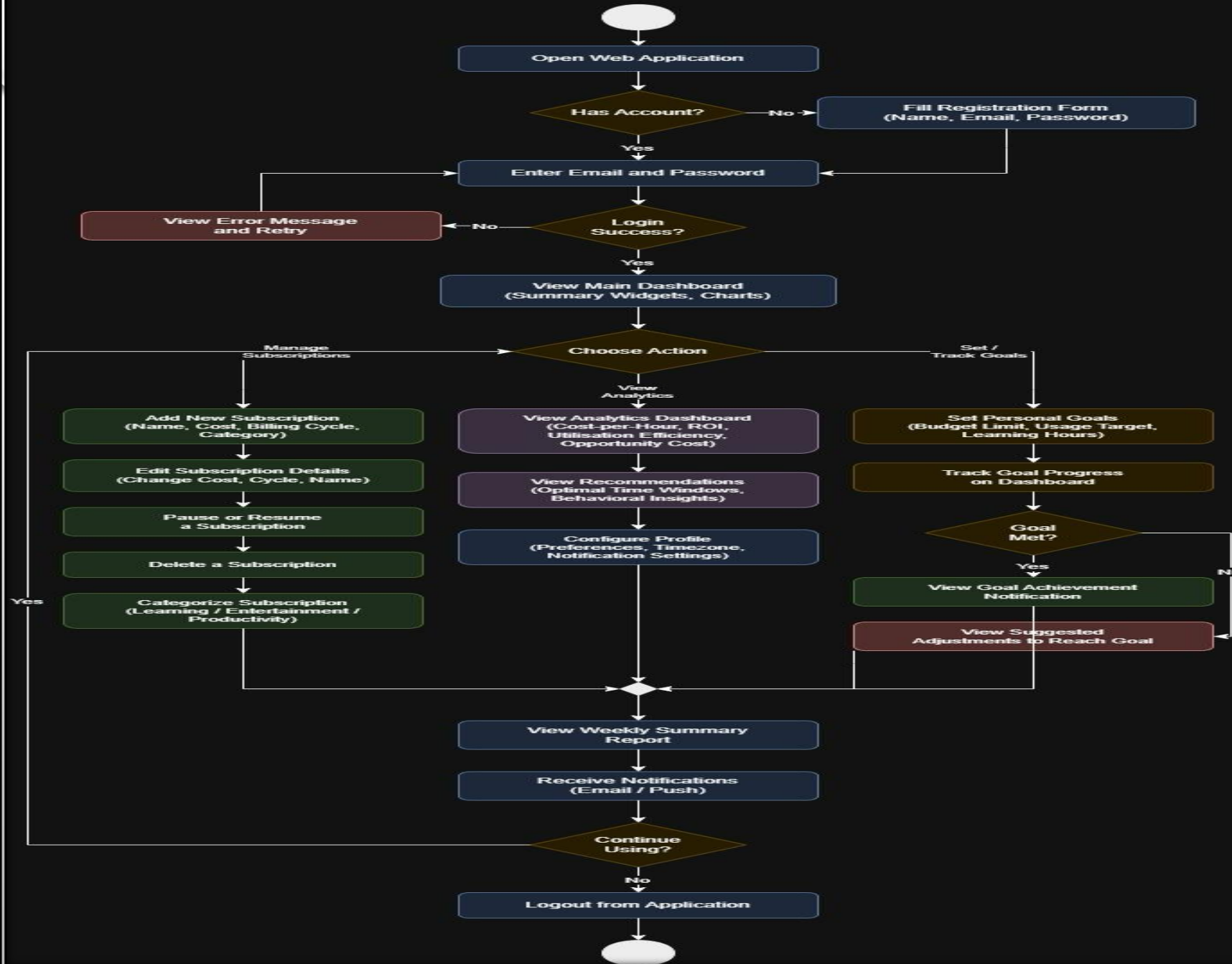


USE CASE MODELING (UML)

SYSTEM WORKFLOW (ACTIVITY DIAGRAM)

- **AUTHENTICATION:** SECURE REGISTRATION AND JWT-BASED LOGIN.
- **DATA INPUT:** USER ADDS FINANCIAL DATA; AGENTS COLLECT DEVICE USAGE DATA.
- **PROCESSING ENGINE:**
 - CALCULATE COST-PER-HOUR FOR EACH SERVICE.
 - TRIGGER REMINDERS FOR UPCOMING BILLS.
- **OUTPUT:** INTERACTIVE DASHBOARD WITH VISUALIZATION (CHART.JS) AND OPTIMIZATION SUGGESTIONS.

Activity Diagram — End User Flow



SYSTEM WORKFLOW (ACTIVITY DIAGRAM)

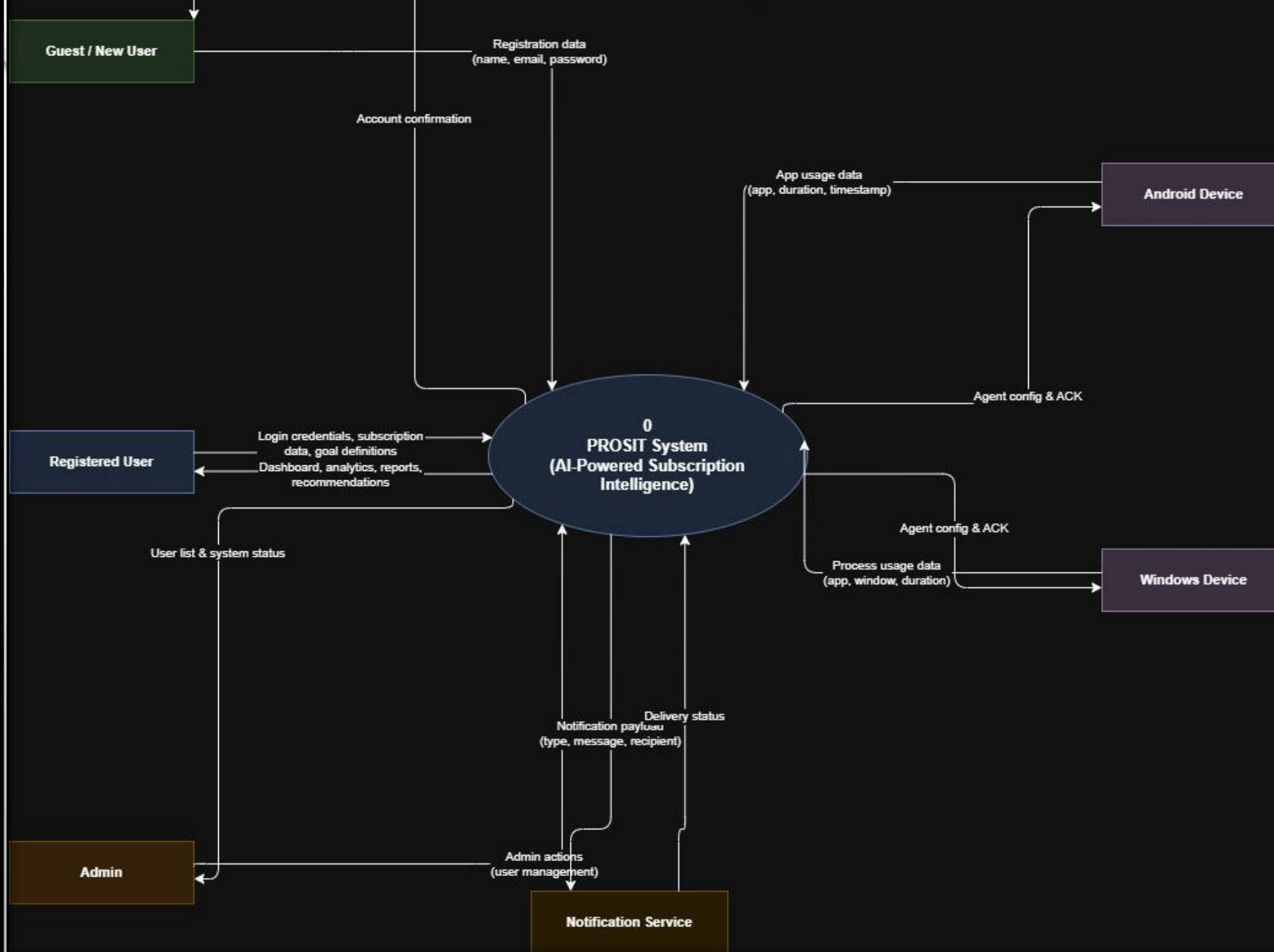
LEVEL 0 (CONTEXT): ILLUSTRATES THE HIGH-LEVEL DATA EXCHANGE BETWEEN THE USER, EXTERNAL DEVICES (AGENTS), AND THE PROSIT SYSTEM.

LEVEL 1 (FUNCTIONAL DECOMPOSITION):

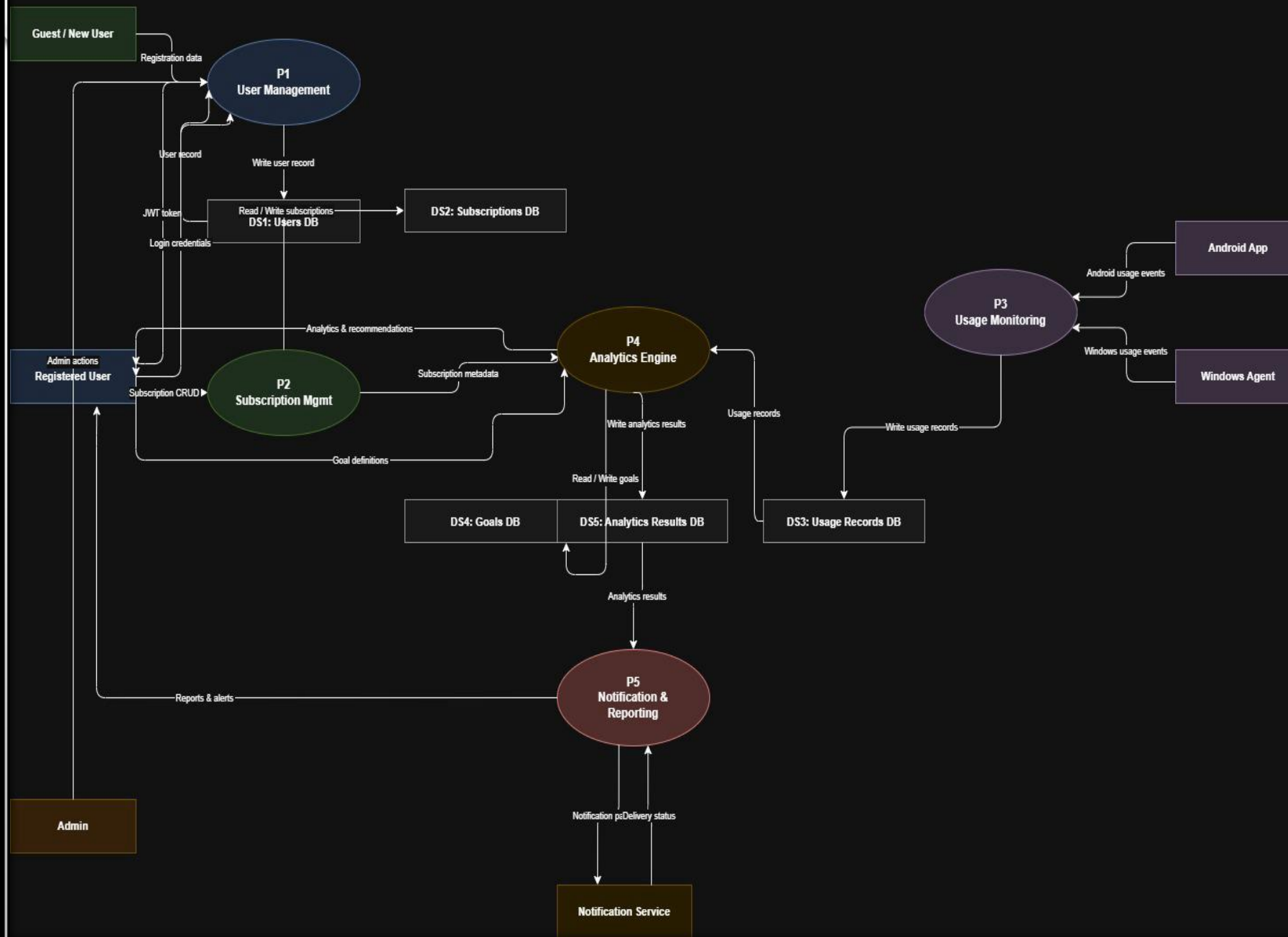
- **P1 - USER MANAGEMENT:** HANDLES PROFILES AND SECURITY.
- **P2 - SUBSCRIPTION MANAGEMENT:** PROCESSES BILL CYCLES AND CATEGORIES.
- **P3 - USAGE MONITORING:** AGGREGATES BACKGROUND DATA FROM AGENTS.
- **P4 - ANALYTICS ENGINE:** THE "BRAIN" – COMPUTES FINANCIAL TRENDS AND ROI.

DATA FLOW ANALYSIS (DFD LEVEL 0 & 1)

DFD Level 0 — Context Diagram

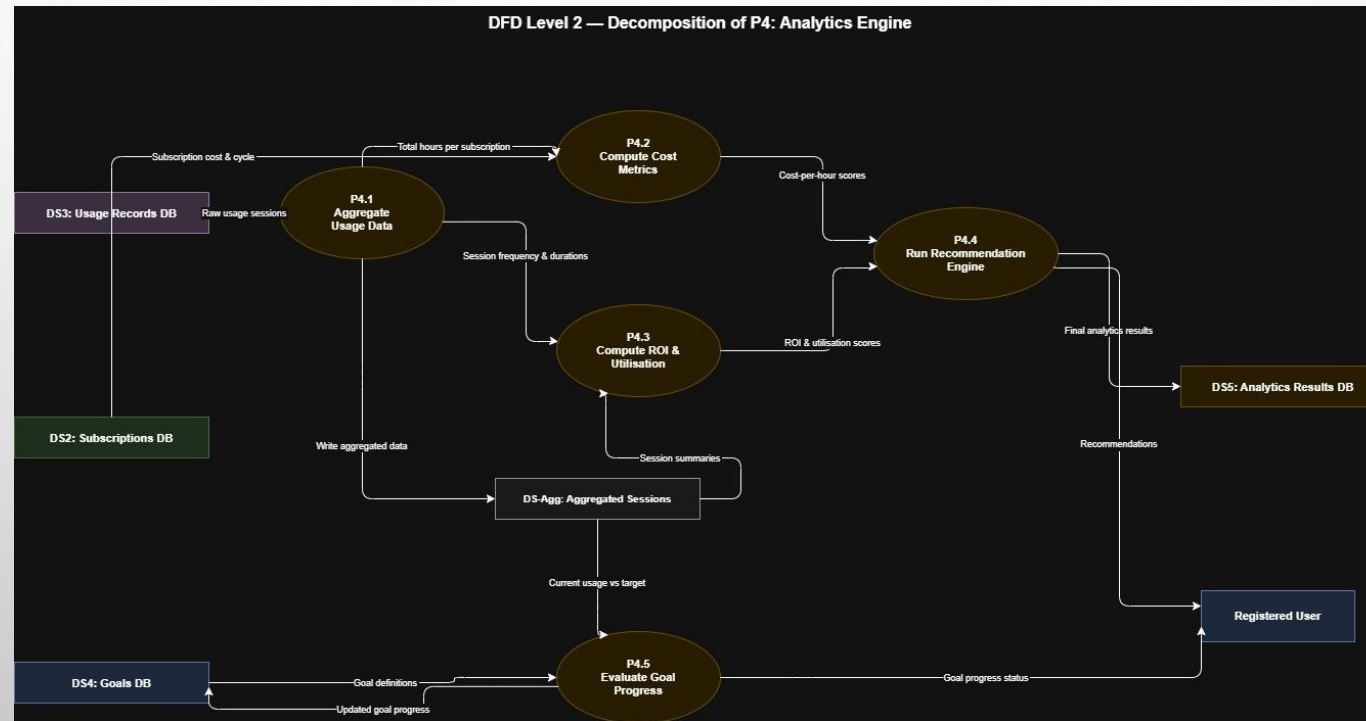


DATA FLOW ANALYSIS (DFD LEVEL 0)



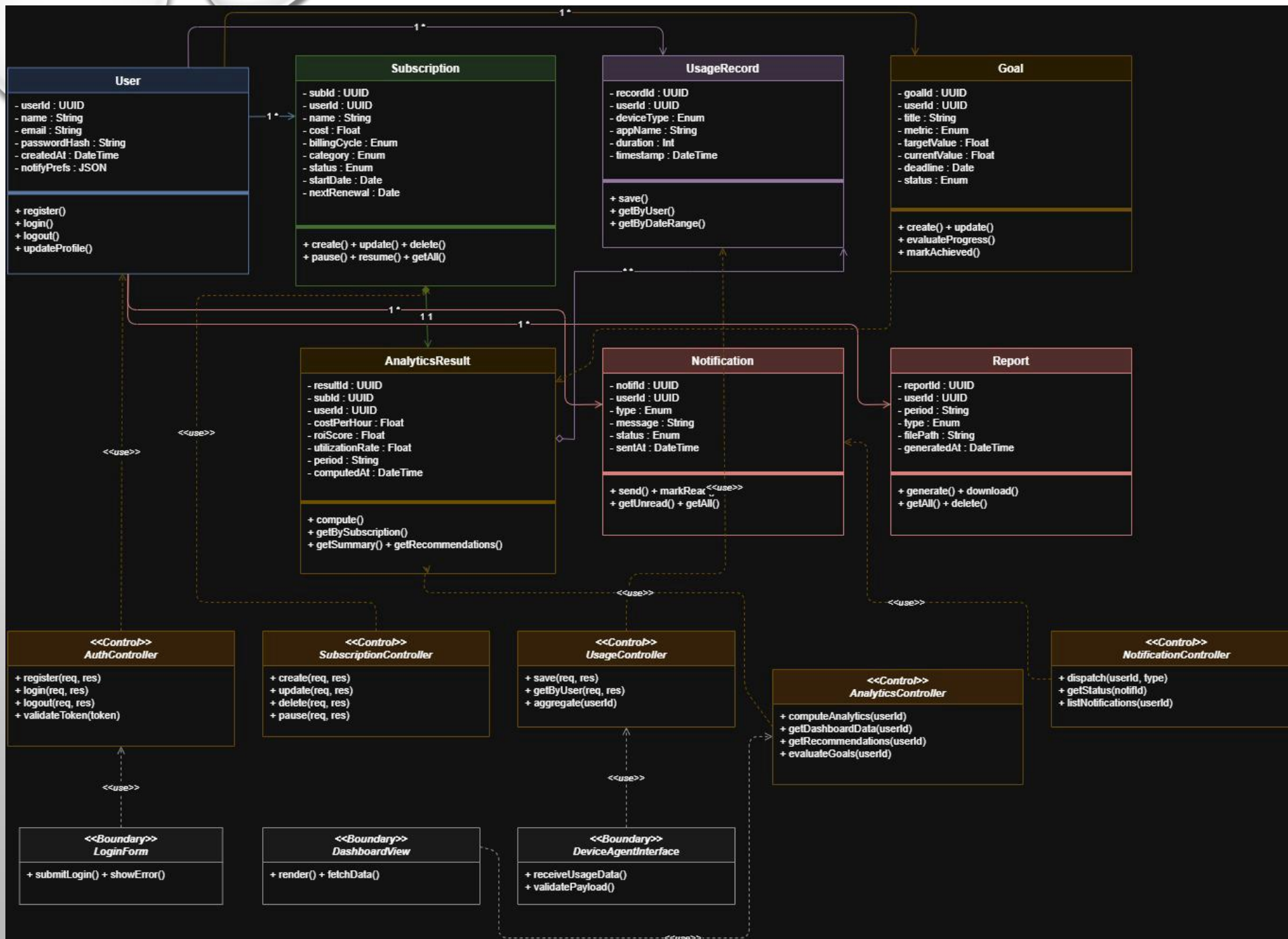
DATA FLOW ANALYSIS (DFD LEVEL 1)

DFD LEVEL - 2



PROPOSED PRODUCT DESIGN (CLASS DIAGRAM)

- **ARCHITECTURE:** THREE-LAYER MVC-ALIGNED DESIGN (ENTITY, BOUNDARY, CONTROL).
- **ENTITY LAYER:** MAPS TO DATABASE (USER, SUBSCRIPTION, USAGERECORD, GOAL).
- **CONTROL LAYER:** BUSINESS LOGIC (ANALYTICSCONTROLLER, NOTIFICATIONCONTROLLER).
- **BOUNDARY LAYER:** INTERFACES FOR WEB UI AND DEVICE AGENTS.
- **KEY LOGIC:** SUBSCRIPTION AND ANALYTICSRESULT SHARE A **COMPOSITION** RELATIONSHIP—RESULTS ARE INTRINSICALLY TIED TO THE SUBSCRIPTION LIFE CYCLE.



PROPOSED PRODUCT DESIGN (CLASS DIAGRAM)

SPRINT 1 PROGRESS & ACHIEVEMENTS

- **REQUIREMENT ENGINEERING:** COMPLETED IDENTIFICATION OF FUNCTIONAL/NON-FUNCTIONAL REQUIREMENTS AND ACTOR MAPPING.
- **SYSTEM MODELING:** FINALIZED ALL CORE UML DIAGRAMS (USE CASE, ACTIVITY, CLASS).
- **ARCHITECTURAL DESIGN:** SUCCESSFULLY DEFINED THE THREE-LAYER SYSTEM ARCHITECTURE FOR MAXIMUM SCALABILITY.
- **RESEARCH:** FINALIZED TECHNOLOGY STACK SELECTION AND AGENT COMMUNICATION PROTOCOLS.

FUTURE ROADMAP & CONCLUSION

- **IMMEDIATE NEXT STEPS (SPRINT 2):**

- DEVELOPMENT OF THE BACKEND API CORE.

PROTOTYPING THE WINDOWS/ANDROID BACKGROUND USAGE TRACKERS.

- **ADVANCED FEATURES:** IMPLEMENTING "WHAT-IF" SIMULATIONS FOR BUDGET PLANNING.
- **SUMMARY:** PROSIT AIMS TO TURN PASSIVE FINANCIAL TRACKING INTO ACTIVE RESOURCE OPTIMIZATION, ENSURING USERS SPEND MONEY ONLY ON SERVICES THEY TRULY VALUE.